UCLA Engineering
HENRY SAMUELI SCHOOL OF ENGINEERING AND APPLIED SCIENCE
Mechanical and Aerospace Engineering Department

January 27, 2026
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Dear Members of the Selection Committee,

I am pleased to write this letter in strong support of the nomination of Parnian Hemmati for the Distinguished Teaching Assistant Award on behalf of the Mechanical and Aerospace Engineering academic unit. Parnian has served as a Teaching Assistant for MECH&AE 103 (Elementary Fluid Mechanics) across multiple quarters over the past three academic years, and she currently holds the role of Teaching Fellow. I served as the instructor of record for this course during her TA appointments and had direct, sustained opportunities to observe her teaching and mentorship.

In my twenty-three years as a faculty member at UCLA, during which I have worked with a large number of graduate Teaching Assistants across multiple courses and academic levels, I can state unequivocally that Parnian Hemmati is the best Teaching Assistant I have had. This assessment is not made lightly. It reflects both the consistency of her instructional excellence and the depth of her impact on student learning. While many TAs perform their assigned duties competently, Parnian operates at a distinctly higher level, demonstrating pedagogical maturity, intellectual clarity, and a level of responsibility typically associated with much more senior educators.

Parnian demonstrates a sustained pattern of evidence-based, student-centered teaching. She addresses students' challenges through intentionally structured discussion sections that emphasize conceptual understanding, and active engagement. In her sections, Parnian encourages students to articulate their problem-solving approach before solutions are presented. She creates space for students to identify relevant equations and engage critically with the material. I observed that this instructional strategy led to higher levels of participation and deeper engagement across discussion sections.

Student feedback across multiple terms consistently highlights Parnian's clarity, organization, approachability, and genuine concern for student learning. She has maintained consistently exceptional overall TA evaluation ratings across repeated appointments, reflecting both the effectiveness and consistency of her teaching. Students frequently note that her discussion sections were instrumental to their understanding of course material and their performance on exams.

What truly distinguishes Parnian, however, is that her contributions go well beyond those of a top Teaching Assistant in the conventional sense. She does not view teaching as a transactional responsibility, but as a personal commitment to student growth and success. She consistently takes the

initiative to support students who are struggling, uncertain, or disengaged, and she does so with patience, empathy, and professionalism.

Parnian is highly approachable during office hours and frequently goes beyond addressing individual homework questions, taking time to connect course material to broader applications in engineering and research. Many students seek her guidance not only for course-related challenges but also for advice regarding academic pathways and future opportunities, reflecting the trust and respect she has earned as an educator.

Parnian's dedication and reflective approach to teaching have had a clear and lasting positive impact on the undergraduate learning experience in our department. She represents the Mechanical and Aerospace Engineering academic unit exceptionally well. I recommend Parnian Hemmati for the Distinguished Teaching Assistant Award without reservation and with my strongest endorsement.

Sincerely,



H. Pirouz Kavehpour, Ph.D.
Professor of Mechanical & Aerospace Engineering
Professor of Bioengineering
Professor of Neurosurgery
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Over the past three years, I have served across multiple quarters as a Teaching Assistant for MECH&AE 103 (Elementary Fluid Mechanics), a large-enrollment, mathematically intensive core undergraduate course that many students find abstract and intimidating. Through repeated appointments to this course, most recently as a Teaching Fellow, I have developed an intentional, evidence-based teaching approach focused on removing barriers to learning, building confidence in problem solving, and fostering a supportive classroom environment.

Across multiple quarters, I observed consistent challenges faced by students. Many struggled to translate lectures into effective problem-solving strategies and were unsure how to begin unfamiliar problems. Others relied on memorizing equations without understanding their physical meaning, leading to anxiety and disengagement. During discussion sections, students often fell behind while copying complex figures and derivations, which detracted from conceptual understanding. These challenges disproportionately affected students who already felt hesitant or underrepresented in technical learning spaces.

To address these barriers, I designed my discussion sections around a highly structured and consistent format that supports cognitive scaffolding and active learning. Each session follows a predictable flow: a concise conceptual review emphasizing physical intuition, followed by guided practice problems aligned with common misconceptions and exam-relevant scenarios (Artifact 1). This structure reduces anxiety, helps students orient themselves cognitively, and provides a clear framework for approaching complex problems.

A central feature of my teaching is intentional guided think-time. After introducing each problem, I give students five to seven minutes to work independently before any solution is presented. During this low-stakes period, students outline solution strategies and identify governing principles. We then solve the problem together, discussing multiple approaches and common pitfalls. During collective problem-solving, I intentionally highlight and credit solution strategies shared by students, emphasizing reasoning over final correctness and encouraging broader participation (Artifact 1).

I observed that students often struggled to keep up during discussions, and I adjusted my teaching approach to address this. Students often lost momentum while redrawing figures during discussions. In response, I began uploading annotated figures and structured notes to the Bruin Learn site prior to each section, allowing students to focus on conceptual understanding rather than transcription (Artifact 2). This change significantly improved pacing, engagement, and the depth of discussion.

I address students' perception of fluid mechanics as abstract by anchoring theoretical concepts in real-world engineering systems. I incorporate short videos and case studies from aerospace, biomedical, and environmental applications, as well as examples from my own research, to illustrate how governing equations describe real physical systems in modern engineering practice (Artifact 3). Beginning with recognizable phenomena helps students develop physical intuition and see the relevance of course material beyond the classroom.

Student feedback provides strong evidence of the effectiveness of these strategies. Across multiple quarters, students consistently describe discussion sections as "essential," "instrumental," and

“where the most learning happened,” highlighting the clarity, organization, and usefulness of the sessions. (Artifact 4).

Creating an inclusive and supportive learning environment is central to my teaching. I foster a non-judgmental classroom culture where questions are welcomed, mistakes are treated as part of the learning process, and struggle is normalized. Students frequently describe my discussion sections as approachable and supportive, noting that this environment makes them more comfortable participating and asking questions (Artifact 4). Outside of class, I hold accessible office hours and provide individualized academic support. Several students, particularly women and those from underrepresented backgrounds, have sought guidance on coursework, research opportunities, and graduate school pathways, with some subsequently joining my research lab or expressing renewed confidence in pursuing advanced engineering degrees.

In addition to my instructional responsibilities, I contribute to teaching development within the department by mentoring other Teaching Assistants. I trained a junior TA newly assigned to MAE 103 by sharing structured discussion materials, example problems, and evidence-based instructional strategies developed over multiple quarters. Through regular meetings and feedback, I supported his development as an instructor. A peer testimonial documents this mentoring process and its impact (Artifact 5).

My teaching is grounded in intentional instructional design, evidence-based strategies, and a commitment to inclusive learning. By identifying barriers to student learning and implementing targeted interventions, I aim to support both academic success and students’ confidence in their identity as engineers. The impact of these practices is reflected in consistently strong student evaluations across eight quarters of teaching, with an average overall rating of approximately 8.5 out of 9 and recurring qualitative feedback highlighting clarity, organization, and approachability (Artifact 4). I am committed to continual reflection and improvement and to contributing to a supportive and effective learning environment at UCLA.

I believe knowledge is power. Not as a slogan, but as a lived experience. As a female engineer from Iran, education gave me agency: the confidence to articulate my ideas, the courage to challenge myself intellectually, and the resolve to move to another country to pursue a PhD in mechanical engineering. Through learning, I gained not only technical expertise, but also the ability to navigate new environments, engage with diverse perspectives, and uplift others along the way. Because education has shaped my life so profoundly, I am deeply motivated to share that power with others through teaching. That experience shapes not only why I value education, but how I think about teaching.

Guided by this perspective, my teaching philosophy is grounded in the belief that effective learning occurs when students are actively engaged, supported through intentional structure, and encouraged to build confidence in their own thinking. In mathematically intensive engineering courses, students often equate success with speed rather than understanding. My goal is to help students shift from passive reception of information to active problem solving, while fostering an inclusive environment in which all students feel capable of succeeding. I view my role not as delivering answers, but as guiding students toward developing their own problem-solving frameworks.

I see structure as a powerful tool for learning. Many students struggle not because they lack ability, but because they lack a clear framework for approaching complex problems. In my teaching, I prioritize consistent and predictable instructional design that helps students orient themselves cognitively. I organize instruction around clear learning goals and a deliberate progression from physical intuition to mathematical formulation, emphasizing how equations emerge from underlying principles rather than presenting them as isolated steps.

Active learning is central to my approach. I intentionally create opportunities for students to think, attempt, and reflect before being shown complete solutions. By pausing for conceptual questions and encouraging students to articulate their reasoning, I promote deeper engagement and collective sense-making. Helping students “feel” the physics behind equations supports understanding that is more durable and transferable than memorization, preparing them to apply foundational principles across new problems in coursework, research, and industry.

Creating a supportive and inclusive learning environment is a priority in my teaching. Engineering classrooms can be intimidating, and many students hesitate to participate out of fear of being wrong. I work intentionally to normalize uncertainty and treat mistakes as a natural part of the learning process. When students offer incomplete or incorrect responses, I respond with curiosity and guidance, helping the class arrive at correct reasoning together while preserving confidence and motivation.

I am particularly mindful of students who may doubt their belonging in engineering spaces. As a woman in a male-dominated field, I understand the importance of representation and mentorship. I strive to be approachable and transparent, often sharing my own academic challenges when students face similar struggles. By validating effort and diverse learning paths, I aim to help students build confidence in their intellectual identity and trust their ability to succeed.

Beyond the classroom, I view teaching as extending into mentorship. I encourage students to apply what they learn to real-world problems and future opportunities, and I actively support those interested in research or graduate study. Helping students envision new academic paths and recognize their potential as engineers is one of the most rewarding aspects of my role.

I view teaching as a reflective and evolving practice. Each class offers an opportunity to refine my approach and better support student learning. My long-term goal is to continue developing teaching practices that empower students to think critically, engage creatively with complex problems, and see themselves as capable engineers and scientists. Whether in the classroom, laboratory, or mentoring relationship, I am committed to fostering learning environments that are rigorous, inclusive, and supportive.

Artifact 1: Annotated Discussion Notes Demonstrating Structured, Active Problem-Solving

Description:
Annotated excerpts from discussion notes showing conceptual review, guided think-time, structured problem-solving steps, and student-contributed solutions.

Conceptual Review. Brief review of governing equation, physical meaning of terms, and assumptions before problem-solving.

Bernoulli's equation:

$P + \frac{1}{2}\rho V^2 + \rho g z = \text{Constant}$ along a streamline

Assumptions:
1. Steady → not changing with time
2. Inviscid → no viscous forces
3. Incompressible → Density not changing



3.32 The tank shown in Fig. P3.32 contains air at atmospheric pressure above the water surface. The velocity of the water flowing from the tank is 22.5 ft/sec. Determine the water level (h).

Practice Problem. Problems are carefully selected to prepare students for homework and exams.

Guided Think Time. Students work independently for 5–7 minutes to identify points, assumptions, and solution strategy before discussion.

Structured Problem-Solving Framework. Problems are approached through consistent steps.

1) Select the points
2) write the assumptions
3) choose the equation
4) Solve for parameters

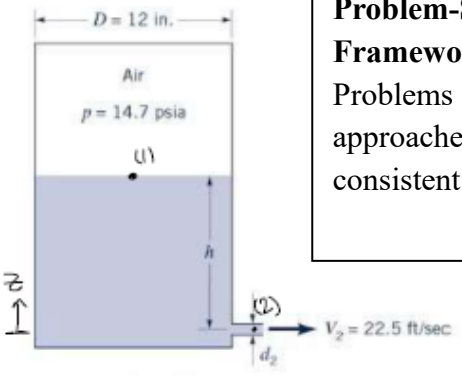


Figure P3.32

$h_{kyle} = 7.8$ ft

Adam

Bernoulli Point 1, 2 $\quad \frac{P_1}{\rho} + \frac{1}{2}V_1^2 + gz_1 = \frac{P_2}{\rho} + \frac{1}{2}V_2^2 + gz_2$ ← Nicky

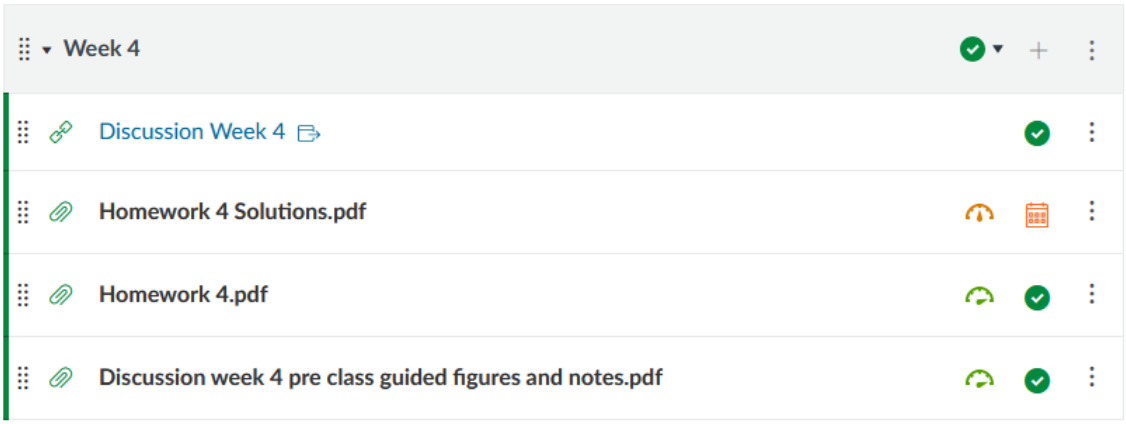
(P_1, P_2 = Patm; $z_2 = 0$)

$\Rightarrow \rho g h = \frac{1}{2}\rho V_2^2 \rightsquigarrow h = \frac{V_2^2}{2g} \longrightarrow h = \frac{(22.5)^2}{2 \times 32.2} = 7.86$ ft

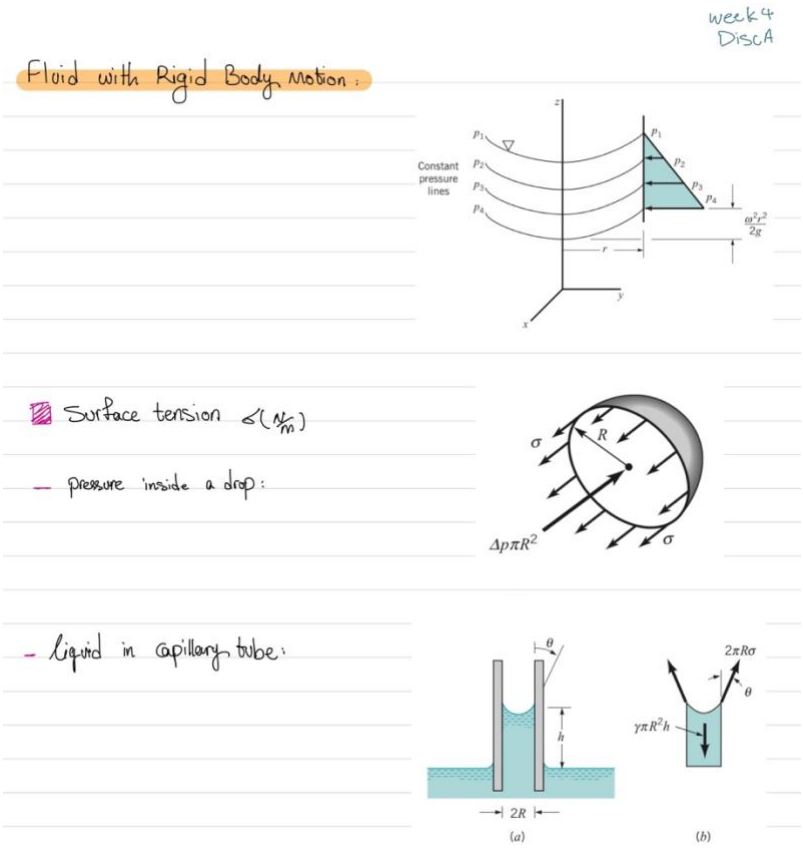
Student Contributions Highlighted. Solution steps and approaches are labeled with student names to recognize participation and encourage engagement.

Artifact 2: Pre-Class Figures and Guided Notes Uploaded to Bruin Learn

Description:
Screenshots and excerpts from Bruin Learn weekly modules demonstrating the use of pre-class figures and guided notes uploaded prior to discussion sections. These materials allow students to focus on conceptual understanding and problem-solving during class rather than transcription.



Screenshot of Bruin Learn weekly module showing pre-class discussion materials uploaded prior to section.

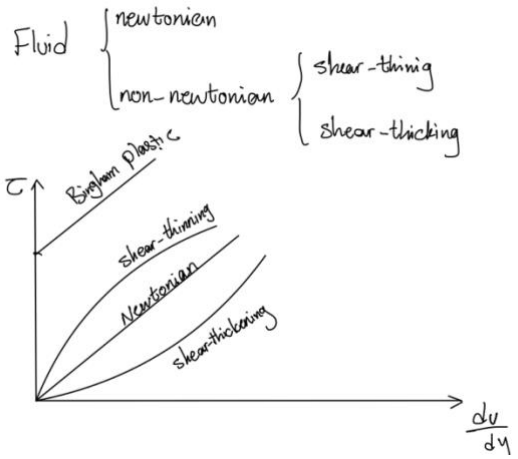


Excerpt from *Discussion week 4 pre class guided figures and notes.pdf*.

Artifact 3: Real-World and Research-Based Examples Connecting Theory to Engineering Practice

Description:
Use of real-world engineering systems and research-based examples to connect fluid mechanics theory to physical applications and build student intuition.

Notes from class

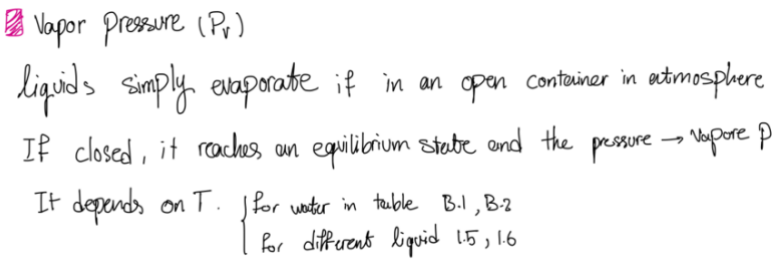


Screenshot of Video shown to connect theory to real life



Non-Newtonian Fluid Behavior: Video demonstration using a cornstarch suspension to introduce shear-thickening behavior at the start of the topic.as a new topic

Notes from class



Example from my research to show the use of theory in research



Vapor Pressure and Cavitation: Research-based example illustrating cavitation phenomena in biological systems to connect vapor pressure concepts to real engineering and biomedical applications.

Artifact 4: Summary of Student Feedback and Teaching Evaluations

Description:
Selected anonymized student comments and overall teaching evaluations across multiple quarters, highlighting clarity, organization, approachability, and instructional impact.

Selected Anonymized Student Feedback

MAE 103 – Elementary Fluid Mechanics (Discussion Sections)
Parnian Hemmati

Class Structure & Organization

"Very organized presentation with a good format for each discussion (recap of topic, then practice problems)."
"Excellent TA. Everything was organized and very thoroughly explained."
"The structure of her discussion sections where she first reviewed the topics of the week, then proceeded to examples was extremely helpful."
"Parnian is able to answer all our questions, summarizes the weeks learnings then provides helpful examples."
"Discussion summed up everything pretty well and comprehensively."
"Her discussion section was essential to go to. I always felt like both the review and the practice problems were where the most learning happened for me in this course."
"Practice problems in discussion sessions were extremely helpful for homework and exams."
"The practice problems were also very helpful, and I appreciated how she made sure she finished as much as she could."
"Discussion sections were instrumental for my learning of all of the content in this course."

Knowledge & Ability to Explain

"The teaching assistant was very knowledgeable and helpful with course material and made discussion sections very worthwhile."
"TA had very good understanding of the material and made sure to cover anything we might have missed while watching the long videos."
"Hemmati was an excellent TA, she was very good at providing helpful explanations that clarified even the more difficult topics."

Clarity & Conceptual Understanding

"Very good at explaining concepts to better understand and always willing to help the students."
"She is very good at teaching and has very clear notes. She is also very knowledgeable when asked questions."
"She explained the topics was helpful and very clear, and she answered all of our questions in a way that made sense."

Engagement & Motivation

"I felt like I was actually learning the course material through her."
"Discussion was absolutely invaluable for applying the concepts from lecture to problems."
"She completely carried the course without her I would have really struggled."

“I learned the majority of material from her discussions.”
“Her support made a difficult course manageable.”
“I wish I could have her for more courses in the future.”

Approachability & Support

“She is super helpful! Her discussions really solidified my understanding of course material, and she was always super accommodating and approachable if we ever had any questions.”
“Office hours were also amazing and very helpful too. I always felt welcome asking whatever questions I had.”
“She always made sure that my question was a ‘good question’ and did not judge me if the questions were obvious.”
“One of the most approachable TAs I have ever had.”
“She was very understanding of everyone’s level in the class which allowed her to explain the topics in a way in which we would all understand.”
“Her discussion sections always had a good vibe and more participation than a usual engineering discussion.”
“I owe this TA my grade!”
“I would not have been successful in this course without her help.”
“The TA was the highlight of this course.”

Summary of Overall Teaching Assistance Evaluations

(Scale: 1–9)

Quarter	Overall Rating
Fall 2023	8.5
Winter 2024	8.8
Spring 2024	8.2
Fall 2024	8.3
Winter 2025	8.4
Spring 2025	8.8
Summer 2025	8.2
Fall 2025	8.4

Overall ratings of TA across eight quarters of MECH&AE 103 discussion sections.

Artifact 5: Peer Testimonial Documenting TA Mentorship and Instructional Leadership

Description: Testimonial from a co-Teaching Assistant describing mentorship, instructional support, shared materials, and pedagogical guidance provided during a first TA appointment.

To whom it may concern,

My name is Edoardo Leone, and I am Parnian's lab colleague, as well as her co-TA for MECH&AE 103 for Fall '25. I am writing to share my experience working alongside Parnian.

Last quarter was my first quarter as a TA, and Parnian's mentorship was key for me to hit the ground running. At the beginning of the quarter, she personally took me to book a time slot for my office hours, and she showed me how to print materials for the class. On top of all this logistical help, she also helped me fine-tune my TA soft skills. She talked me through a number of ways in which she likes to make sure that students feel encouraged to ask for help. Thanks to her tips, I was able to appear relatable and approachable to my section, which was reflected in high discussion and OH attendance throughout the quarter, and exclusively positive reviews on the Students' Experiences of Teaching survey.

Parnian also provided me with materials she had prepared for the class discussions. I estimate that her materials saved me 4-5 hours of work per week. Her materials are extremely high in quality and easy to understand, as she includes side notes together with her numerical solutions so that every step is clear for the students. Furthermore, when grading exams, Parnian always went out of her way to double-check my solutions and mark schemes, making sure that I was being fair to the students.

For all the reasons described above, Parnian is an indefatigable worker, but I also want to highlight her adaptability as a TA. Despite having a 10-12 am discussion, she agreed to cover my 8-10 am one during a week when I could not attend. That meant commuting earlier to campus, giving up an entire morning of research work, repeating the same material twice in a row, and working uninterruptedly for 4 hours.

In a male-dominated field such as Mechanical Engineering, Parnian leads the female students in our class by example. After she shared details about her research project during class discussions, we had multiple female students from Parnian's section reach out to our lab inquiring about open positions. To me, this shows that she conveys her passion for her research to her students and she instills confidence in them. I aspire to be able to do that, too, one day. Having been an undergraduate student at UCLA myself, I know what a TA who goes above and beyond looks like. And Parnian definitely falls in this category. I can therefore say with certainty that awarding her as a Distinguished Teaching Assistant would be a rightful application of the laws of meritocracy.

01/28/2026



Edoardo G. Leone

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name

Briefly describe how you know the nominee. (50 words max)

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

Distinguished Teaching Assistant — Letter of Recommendation

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January 27, 2026

Dear Selection Committee:

It is my pleasure to write this letter in strong support of Parnian Hemmati for the UCLA Distinguished Teaching Assistant Award. For more than two years in service as a Teaching Assistant for MAE 103, Elementary Fluid Mechanics, Parnian has demonstrated an exceptional commitment to undergraduate education, pedagogical innovation, and student mentorship.

MAE 103 is widely regarded as a challenging course, as it is often students' first formal exposure to fluid mechanics. Parnian recognized early on that many students struggle not only with the mathematics, but also with developing the physical intuition needed to understand the subject. To address this challenge, she has consistently incorporated real-world engineering examples into discussion sections to contextualize abstract concepts before formal derivations are introduced.

Parnian's research expertise is in the fluid mechanics of the brain. Drawing both from her research experience and from familiar demonstrations, Parnian connects topics such as pressure variation and vapor pressure to real-world examples of brain injury mechanics, she has introduced compressible flow through accessible aerospace applications, and she has used engaging examples—such as non-Newtonian behavior illustrated through corn-starch pool demonstrations—to bring complex ideas to life. These examples spark curiosity, encourage student questions, and create natural transitions into the core course material, greatly enhancing student engagement and understanding.

In addition to strengthening conceptual intuition, Parnian has made a deliberate effort to improve students' problem-solving skills. She has observed that many students struggle with how to begin approaching complex problems. To address this, Parnian structures her discussion sections around active problem setup, asking students to spend dedicated time independently or collaboratively to identify relevant equations, assumptions, and solution strategies before working through solutions as a group. This approach has fostered deeper engagement, increased student confidence, and helped students develop transferable problem-solving habits that extended beyond the course itself.

Beyond the classroom, Parnian has been an exceptionally approachable and supportive mentor. Through office hours and informal follow-up conversations, she has provided guidance to students interested in research and graduate study. These efforts have had a tangible and lasting impact: more than ten students have gone on to join her the research lab of her PI (Prof. Pirouz Kavehpour) for undergraduate research after completing the course, a testament to Parnian's ability to inspire and mentor students at a formative stage in their academic careers.

Across multiple quarters, Parnian's dedication and effectiveness have been consistently reflected in outstanding teaching evaluations, with average scores exceeding 8.5 out of 9 and overwhelmingly positive student feedback. Students repeatedly note her clarity, enthusiasm, and genuine investment in their learning.

In summary, Parnian exemplifies the qualities of a Distinguished Teaching Assistant: pedagogical creativity, a student-centered approach to learning, and a deep commitment to mentorship and academic excellence. As a former winner of the Distinguished Teaching Award, I recommend Parnian for this award without reservation.

Kindest regards,



Jeff Eldredge
Professor and Department Chair, Mechanical & Aerospace Engineering